Interstitial Lung Disease 2 (ILD2, SFTPA2): Comprehensive Disease Report

Disease: Interstitial Lung Disease 2 (ILD2) **Causal gene: SFTPA2** (surfactant protein A2; HGNC:10799; UniProt Q8IWL1; chr10q22.3) **MONDO:** MONDO:0800497 · **OMIM:** 178500 · **DOID:** 0060971 · **GARD:** 0028066 · **MedGen/UMLS:** C5561926

Summary

Interstitial Lung Disease 2 (ILD2) is a rare, **autosomal-dominant, adult-onset form of familial pulmonary fibrosis caused by heterozygous missense mutations in SFTPA2**, the gene encoding surfactant protein A2 (SP-A2). It must not be confused with similarly named interstitial lung diseases driven by other genes (e.g., SFTPB-, SFTPC-, ABCA3-, NKX2-1-, or telomere-gene–related fibrosis); the defining lesion of ILD2 is a coding mutation in **SFTPA2**, most characteristically in the C-terminal **carbohydrate-recognition domain (CRD)** of the protein (canonical variants p.Gly231Val and p.Phe198Ser). The disease was originally defined by linkage in a large kindred co-segregating usual interstitial pneumonia (UIP)–pattern pulmonary fibrosis with lung adenocarcinoma, establishing ILD2 as a fibrosis syndrome with a strong cancer predisposition.

The core pathophysiology is a **toxic gain-of-function / proteostasis mechanism** rather than simple haploinsufficiency. Mutant SP-A2 misfolds, is retained in the endoplasmic reticulum (ER) of alveolar type II (AT2) epithelial cells, and fails to be secreted. ER retention activates the **unfolded protein response (UPR)**, which in turn drives secretion of **latent TGF-β1**, autocrine/paracrine TGF-β signaling, **epithelial–mesenchymal transition (EMT)** and epithelial cell death. This cascade produces progressive fibrotic remodeling of the lung with a UIP histologic pattern. The same secretory failure is shared across pathogenic SFTPA1/SFTPA2 variants, and the accompanying bronchiolar metaplasia and epithelial dysregulation are thought to underlie the elevated lung-cancer risk in variant carriers.

Clinically, ILD2 presents earlier than sporadic IPF (familial pulmonary fibrosis median diagnosis ~61 years vs ~67 years, with cases reported from the late teens to 90s), with exertional dyspnea, cough, reduced diffusing capacity, ground-glass opacities and cysts on CT,

and a UIP pattern. Prognosis is poor — surfactant-related-gene (SRG) carriers have low transplant-free survival (median ~44 months in one cohort). There is **no SFTPA2-specific therapy**; management relies on antifibrotics (pirfenidone, nintedanib), selective immunomodulation in some surfactant-gene cases, lung-cancer CT surveillance, genetic counseling, and lung transplantation. This report synthesizes nine confirmed findings drawn from 27 reviewed papers and authoritative database annotations (OMIM, ClinVar, UniProt/InterPro, Monarch/HPO).

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1. Disease Information

Overview. ILD2 is a Mendelian interstitial lung disease defined by Monarch as *"An interstitial lung disease in which the cause of the disease is a variation in the SFTPA2 gene."* It manifests as adult-onset familial pulmonary fibrosis, usually with a **usual interstitial pneumonia (UIP)** histologic pattern, and carries a predisposition to **lung adenocarcinoma**. It falls within the broader clinical category of familial pulmonary fibrosis / familial interstitial pneumonia and is one of the "surfactant-related gene" (SRG) forms of pulmonary fibrosis (distinct from telomere-related gene forms).

Key identifiers.

Resource	Identifier
OMIM	178500
MONDO	MONDO:0800497
DOID	DOID:0060971
GARD	0028066
MedGen / UMLS	C5561926
Gene (HGNC)	SFTPA2 (HGNC:10799)
UniProt	Q8IWL1 (SFPA2_HUMAN)

Synonyms / alternative names. Interstitial lung disease 2; fibrocystic pulmonary dysplasia; cryptogenic fibrosing alveolitis; familial idiopathic pulmonary fibrosis; usual interstitial pneumonitis; idiopathic pulmonary fibrosis (when caused by SFTPA2). Note that several of these synonyms are generic IPF terms; the ILD2 label specifically denotes the SFTPA2-caused subtype.

Nature of the information. The knowledge base here is derived predominantly from **aggregated, disease-level resources** (OMIM, ClinVar, UniProt, Monarch/HPO) and from **published family/cohort studies** rather than from individual EHR records. Evidence sources span human clinical cohorts, in vitro functional assays, and database curation.

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2. Etiology

Primary cause (genetic). ILD2 is caused by **heterozygous germline missense mutations in SFTPA2**, inherited in an autosomal-dominant manner. The original disease-defining study mapped linkage in a large IPF/lung-adenocarcinoma family to chromosome 10 (a 15.7-Mb interval) and identified rare missense mutations **G231V** and **F198S** in SFTPA2, both affecting invariant residues of the highly conserved CRD (PMID: 19100526). The authors reported: *"Both mutations involve invariant residues in the highly conserved carbohydrate-recognition domain of the protein and are predicted to disrupt protein structure. Recombinant proteins carrying these mutations are retained in the endoplasmic reticulum and are not secreted."*

Genetic risk factors. - *Causal variants:* CRD missense variants in SFTPA2 (p.Gly231Val, p.Phe198Ser, and additional ClinVar pathogenic variants such as p.Asn171Ile, p.Tyr186Cys, p.Trp233Arg, and the in-frame p.Tyr240del). - *Shared pathway genes:* Pathogenic variants in the paralog **SFTPA1** cause an essentially indistinguishable SP-A–disorder phenotype, and other surfactant genes (SFTPC, ABCA3, NKX2-1) and telomere-related genes (TERT, TERC, RTEL1, PARN, TINF2, DKC1) cause overlapping familial pulmonary fibrosis. - *Common modifier/ susceptibility loci in broader IPF:* the **MUC5B promoter polymorphism** and SNPs in TERT, FAM13A, and surfactant genes are established IPF risk alleles, though these are general IPF modifiers rather than ILD2-specific (PMID: 35795226).

Environmental risk factors. For SFTPA2-caused disease itself, the mutation is the dominant driver, with incomplete/age-dependent penetrance implying additional modifiers. In the broader IPF context relevant to disease expression, recognized environmental/host risk factors include **aging, male sex, cigarette smoking, and occupational/inhalational exposures**. Smoking is particularly relevant given the lung-cancer predisposition in carriers.

Protective factors. No specific genetic or environmental protective alleles have been established for ILD2. Longer telomere length relative to typical IPF has been noted as a *distinguishing feature* (not necessarily protective) of SFTPA2-variant carriers (PMID: 36135709).

Gene–environment interactions. The incomplete penetrance and variable expressivity of SFTPA2 mutations strongly imply gene–environment and gene–gene interaction (e.g., smoking, aging, and modifier loci shaping whether and when fibrosis or cancer develops), although specific GxE mechanisms have not been formally quantified for ILD2.

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3. Phenotypes

The Monarch Initiative provides an authoritative set of 12 HPO annotations for ILD2 (MONDO:0800497 / OMIM 178500):

Phenotype	HPO term	Type	Notes
Pulmonary fibrosis	HP:0002206	Clinical/imaging/ pathology	Core feature; progressive
Usual interstitial pneumonia	HP:0031950	Histopathology pattern	Predominant pattern in familial PF
Dyspnea	HP:0002094	Symptom	Progressive
Exertional dyspnea	HP:0002875	Symptom	Early, common
Cough	HP:0012735	Symptom	Typically dry/chronic
Clubbing of fingers	HP:0100759	Clinical sign	Sign of chronic hypoxemia/ fibrosis
Decreased DLCO	HP:0045051	Lab/functional	Hallmark of impaired gas transfer
Pulmonary arterial hypertension	HP:0002092	Complication	Secondary to fibrosis/hypoxia
Alveolar cell carcinoma	HP:0006519	Neoplasm	Lung adenocarcinoma predisposition
Elevated BAL neutrophil proportion	HP:0032977	Lab abnormality	BAL finding
Increased circulating immunoglobulin	HP:0010702	Lab abnormality	
Cirrhosis	HP:0001394	Extrapulmonary	More typical of telomere forms; listed in annotation set

Phenotype characteristics. - *Age of onset:* Adult/late-onset; familial pulmonary fibrosis is diagnosed younger than sporadic (mean 61.4 vs 66.6 years, p=0.012), with a wide reported range (19–92 years) (PMID: 22853774). SFTPA2-variant carriers are often younger than typical IPF patients. - *Severity:* Variable but frequently severe; carriers can present with substantially reduced lung function (FVC ~72% predicted, DLCO ~46% predicted in an SRG cohort) (PMID: 36370864). - *Progression:* Progressive and frequently fatal, with low transplant-free survival. - *Frequency:* ILD and/or lung adenocarcinoma was reported in 13/14 families (93%) in a French SP-A disorder cohort (PMID: 32855221).

Quality-of-life impact. Progressive dyspnea, cough, exercise limitation, oxygen dependence, and the psychological burden of a hereditary, cancer-predisposing, often-fatal disease substantially impair daily functioning. Disease-specific QoL instrument values were not resolved in this investigation; this is a knowledge gap.

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4. Genetic / Molecular Information

Causal gene. SFTPA2 (surfactant protein A2), chromosome 10q22.3; encodes SP-A2, an innate-immune collectin of pulmonary surfactant (UniProt Q8IWL1).

Pathogenic variants. Pathogenic/likely-pathogenic variants are predominantly **missense or small in-frame changes clustered in the C-type lectin carbohydrate-recognition domain (CRD)**. ClinVar (queried November 2026) held **137 SFTPA2 records**; among the first 100 reviewed, 5 Pathogenic, 2 Likely pathogenic, 2 Conflicting, ~55 VUS, and ~35 benign/likely benign. Confirmed pathogenic/likely-pathogenic SNVs include (NM_001098668.4):

cDNA	Protein	Type	Domain
c.512A>T	p.Asn171Ile	missense	CRD
c.557A>G	p.Tyr186Cys	missense	CRD
c.697T>A	p.Trp233Arg	missense	CRD
c.719_721del	p.Tyr240del	in-frame del	CRD
(literature)	p.Gly231Val	missense	CRD
(literature)	p.Phe198Ser	missense	CRD

Variant classification/type. Per ACMG/AMP interpretation in ClinVar, the pathogenic set is enriched for missense/in-frame variants in a functionally critical domain; many remaining SFTPA2 records are VUS, reflecting the gene's high natural polymorphism.

Allele frequency. Pathogenic CRD variants are **rare** (private/family-level), consistent with high-penetrance Mendelian disease; functional studies report that deleterious SFTPA2 variants were found at ~1.3% prevalence in an unselected IPF whole-genome cohort (PMID: 36135709).

Somatic vs germline. The disease-causing variants are **germline**. Large 10q duplications classified pathogenic in ClinVar are incidental CNVs, not SFTPA2-specific.

Functional consequence. A **toxic gain-of-function / proteostasis defect**: recombinant mutant SP-A2 proteins are **retained in the ER and not secreted** (PMID: 19100526). Across 11 pathogenic SFTPA1/SFTPA2 mutations, *"protein production was preserved but secretion was abolished"* (PMID: 32855221) — i.e., the lesion is secretory/misfolding rather than loss of expression.

Modifier genes / epigenetics / chromosomal abnormalities. No ILD2-specific modifier genes are firmly established; broader IPF modifiers (MUC5B, TERT, FAM13A) and epigenetic drivers (specific microRNAs/lncRNAs, transposon dysregulation affecting TGF-β/sirtuin/telomere axes) have been proposed for IPF in general (PMID: 35795226). No recurrent chromosomal abnormality defines ILD2.

UniProt/InterPro annotation (Q8IWL1). SP-A2 is a collectin with an N-terminal collagen-like region and a C-terminal C-type lectin CRD (InterPro IPR033990 Collectin_CTLD, IPR001304 C-type lectin; Pfam PF00059 Lectin_C). In the presence of calcium ions it binds surfactant phospholipids and lowers surface tension at the alveolar air–liquid interface, essential for normal respiration. Localization: secreted/surface film, plus lamellar body (GO:0042599), ER membrane (GO:0005789), multivesicular body (GO:0005771), clathrin-coated endocytic vesicle (GO:0045334).

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5. Environmental Information

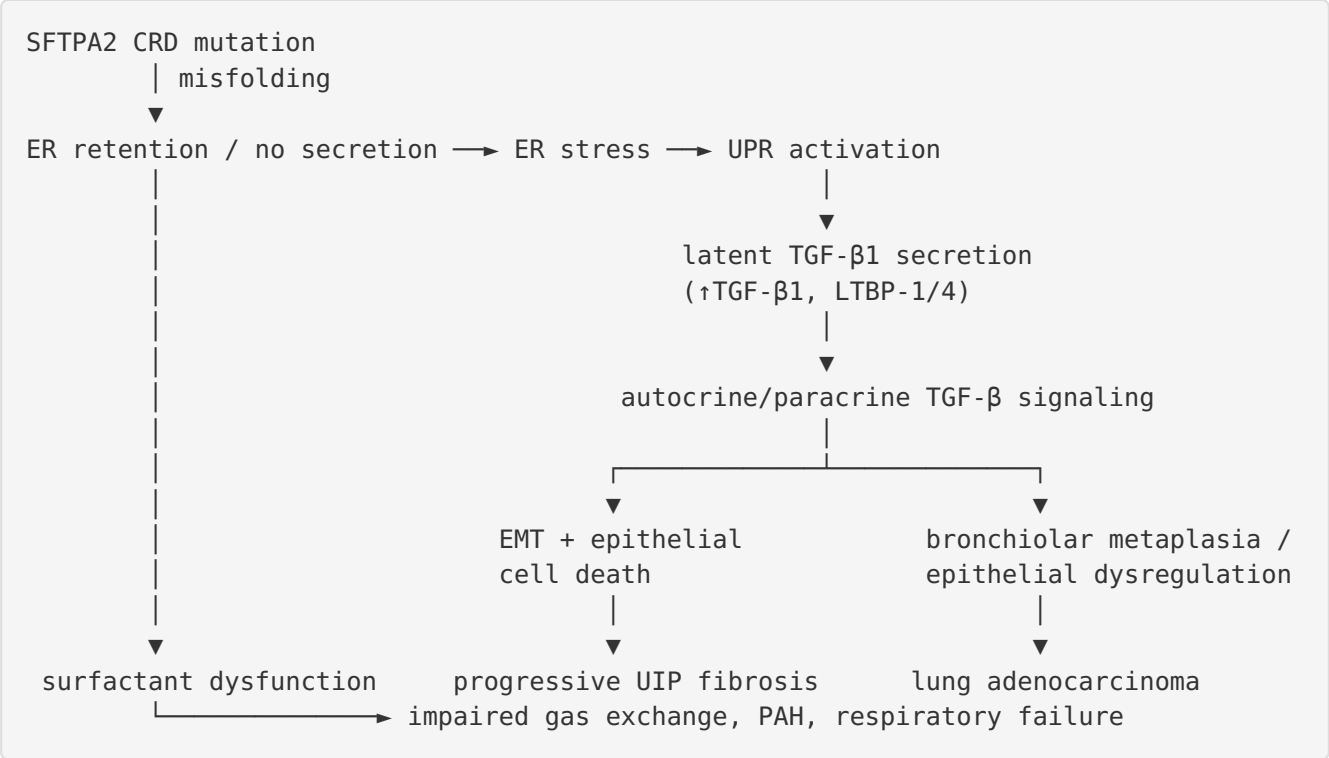
Environmental contributors to ILD2 are those of pulmonary fibrosis generally — **cigarette smoke, inhaled/occupational particulates, and aging** — acting on a genetically primed epithelium. Smoking is doubly relevant given the lung-cancer predisposition. No infectious agent causes ILD2; however, SP-A2's innate-immune role means surfactant dysfunction may alter host defense. There are no established dietary/lifestyle protective exposures specific to ILD2.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **germline missense mutation in SFTPA2** (e.g., p.Gly231Val, p.Phe198Ser) alters an invariant residue of the CRD → **leads to** a misfolded SP-A2 protein (demonstrated: predicted structural disruption; recombinant proteins misfold).
2. Misfolded SP-A2 **is retained in the endoplasmic reticulum** of alveolar type II (AT2) cells and **is not secreted** → protein production is preserved but secretion abolished (demonstrated in recombinant and patient-derived systems).
3. ER accumulation of misfolded protein **results in ER stress and activation of the unfolded protein response (UPR)** (demonstrated; UPR activation required for maximal downstream effect).
4. UPR activation **leads to secretion of latent TGF-β1**, coordinately upregulating TGF-β1, LTBP-1 and LTBP-4 (demonstrated in lung epithelial cells; mutant-specific — not seen with common SP-A2 variants or other misfolded proteins).
5. Latent/active TGF-β1 **drives autocrine and paracrine TGF-β signaling** in the alveolar epithelium (demonstrated).
6. TGF-β signaling **causes epithelial–mesenchymal transition (EMT) and epithelial cell death** (demonstrated; reversed by pan-TGF-β antibody, TGF-β receptor antagonist, or LTBP silencing).
7. Repeated AT2 injury, EMT, and aberrant fibroblast/myofibroblast activation **result in progressive extracellular-matrix deposition and UIP-pattern pulmonary fibrosis** (inferred from integration of steps 1–6 with UIP histology in patients).
8. **Branch:** The same epithelial dysregulation / bronchiolar metaplasia **predisposes to lung adenocarcinoma** (inferred/epidemiologically supported; original families co-segregated fibrosis and adenocarcinoma, and SRG carriers show high cancer risk).
9. Advancing fibrosis **leads to impaired gas exchange (decreased DLCO), exertional dyspnea, hypoxemia, pulmonary arterial hypertension, respiratory failure, and death** (clinically demonstrated).



Detail by category

- **Molecular pathways:** Core driver is **TGF-β1 signaling**, activated via the **UPR/ER-stress** axis; downstream IPF-associated pathways include Wnt/β-catenin and PI3K/Akt (PMID: 42510961).
- **Cellular processes:** Protein misfolding, impaired proteostasis, ER stress, apoptosis/cell death, EMT, and fibroblast-to-myofibroblast differentiation.
- **Protein dysfunction:** Misfolding of the C-type lectin CRD → ER retention → gain-of-function proteotoxicity (not haploinsufficiency).
- **Immune involvement:** SP-A2 is an **innate host-defense collectin** that interacts with the alveolar macrophage and modulates surfactant structure/function (PMID: 34484180); BAL neutrophilia and increased circulating immunoglobulin are observed.
- **Tissue damage mechanism:** Chronic fibrosis with ECM accumulation, driven by repetitive epithelial injury.
- **GO/CL suggestions:** biological processes — response to ER stress (GO:0034976), UPR (GO:0030968), TGF-β receptor signaling (GO:0007179), EMT (GO:0001837), surfactant homeostasis (GO:0043129); cell types — alveolar type II cell (CL:0002063), alveolar macrophage (CL:0000583), (myo)fibroblast (CL:0000057 / CL:0000186).

7. Anatomical Structures Affected

- **Organ level: Lung** (primary; UBERON:0002048), respiratory system (UBERON:0001004). Secondary: pulmonary vasculature → **pulmonary arterial hypertension**; right heart strain (cor pulmonale) as a downstream complication.
- **Tissue/cell level:** Alveolar epithelium (**alveolar type II cells**, CL:0002063) is the primary injured cell; **alveolar macrophages** (CL:0000583) interact with SP-A2; **fibroblasts**/**myofibroblasts** mediate matrix deposition. Fibrosis predominates in the **alveolar interstitium**; bronchiolar metaplasia is noted.
- **Subcellular level: Endoplasmic reticulum** (GO:0005789, site of mutant retention) is central; normal SP-A2 trafficking involves the **lamellar body** (GO:0042599), multivesicular body (GO:0005771), clathrin-coated endocytic vesicle (GO:0045334), and the secreted **surface film** at the air–liquid interface.
- **Localization / lateralization: Bilateral**, typically basal/subpleural-predominant fibrosis consistent with a UIP distribution.

8. Temporal Development

- **Onset:** Adult/late-onset, **insidious and chronic**; familial cases present earlier than sporadic (mean ~61 vs ~67 years; range 19–92) (PMID: 22853774). SFTPA2 carriers skew younger than typical IPF.
- **Progression:** Chronic and **progressive**, often to respiratory failure. SRG-mutation ILD shows low transplant-free survival (**median ~44 months**) (PMID: 36370864); familial IPF shows faster physiological decline in some cohorts (e.g., FVC −2.36%/yr, DLCO −4.44%/yr) (PMID: 41618408).
- **Disease course:** Progressive, lifelong; no spontaneous remission. Partial, temporary stabilization may occur with immunomodulation in some surfactant-gene cases.
- **Critical periods:** Early diagnosis (genetic ascertainment of at-risk relatives) opens a window for surveillance and timely transplant referral.

9. Inheritance and Population

- **Inheritance: Autosomal dominant** (OMIM 178500).
- **Penetrance/expressivity: Incomplete, likely age-dependent penetrance** and **variable expressivity** — some carriers develop fibrosis, some lung cancer, some both, and importantly some SFTPA2 carriers present with IPF **without** a personal/family history of lung cancer (PMID: 36135709).
- **Epidemiology:** ILD2 is rare. Surfactant biology mutations account for roughly **1–3% of IPF**, and functionally deleterious SFTPA2 variants were found in ~1.3% of an unselected IPF genome cohort (PMID: 36135709). Overall IPF prevalence is ~2–29 per 100,000 (PMID: 35795226); ILD2 is a small genetic subset of this.
- **Carrier frequency / founder effects:** Pathogenic variants are rare and often family-private; no established founder variant for SFTPA2 specifically.
- **Demographics:** No strong ethnic predilection established for SFTPA2; Western IPF cohorts show surfactant-gene–driven architecture more than some Asian cohorts (PMID: 41618408). IPF overall is more common in older males and smokers; SRG carriers are younger and less often male than typical IPF (PMID: 36370864).

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10. Diagnostics

Imaging. High-resolution CT is central. Surfactant-gene ILD is characteristically associated with **ground-glass opacities and cysts** (in addition to UIP features): *"Surfactant genes mutations are usually associated with ground-glass opacities and cysts on CT scan..."* (PMID: 32437840).

Functional tests. Pulmonary function testing shows a **restrictive pattern with decreased DLCO** (HP:0045051); carriers may present with markedly reduced FVC (~72% predicted) and DLCO (~46% predicted) (PMID: 36370864).

BAL / labs. Elevated BAL neutrophil proportion (HP:0032977) and increased circulating immunoglobulin (HP:0010702) have been annotated.

Histopathology. UIP pattern predominates in familial pulmonary fibrosis; bronchiolar metaplasia is noted on biopsy/explant in SFTPA2 carriers. Altered lung SP-A expression pattern was observed in 6/28 patients in a functional cohort (PMID: 32855221).

Genetic testing. The definitive diagnostic. Recommended approaches: **NGS gene panels** for familial ILD covering surfactant genes (SFTPA1, SFTPA2, SFTPC, ABCA3, NKX2-1) and telomere genes (TERT, TERC, RTEL1, PARN), **whole-exome/whole-genome sequencing**, or **targeted single-gene testing** of SFTPA2. NGS has uncovered surfactant- and telomere-gene determinants and improves classification of overlapping ILD phenotypes (PMID: 42510961; PMID: 36833253). Karyotyping, FISH, CMA, mtDNA, and repeat-expansion testing are **not applicable**.

Clinical criteria / differential. Diagnosis follows ATS/ERS IPF/ILD frameworks (clinical–radiologic–pathologic integration) plus family history and genetics. Differential includes sporadic IPF, telomere-related PF, SFTPC/ABCA3 disease, hypersensitivity pneumonitis, and connective-tissue-disease ILD. Distinguishing features favoring SFTPA2: younger onset, **longer telomeres** than typical IPF, cysts/ground-glass on CT, and strong family history of fibrosis ± lung adenocarcinoma (PMID: 36135709).

Screening. Cascade genetic testing of first-degree relatives and **lung-cancer CT surveillance** in variant carriers are recommended (PMID: 38575158).

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11. Outcome / Prognosis

Survival. Poor. SRG-mutation ILD carries **median transplant-free survival ~44 months**, with lower baseline FVC and DLCO (PMID: 36370864). Familial IPF can show faster physiological decline than sporadic disease.

Morbidity. Progressive dyspnea, oxygen dependence, exercise limitation, pulmonary arterial hypertension, and the burden of malignancy risk.

Complications. Respiratory failure, PAH/cor pulmonale, acute exacerbations, and **lung adenocarcinoma** (a defining complication of this subtype).

Prognostic factors. Baseline and rate-of-decline in FVC/DLCO, extent of fibrosis, PAH, and development of lung cancer. Lung transplantation can be a valid option with reportedly good post-transplant outcomes in surfactant-disorder patients, though lung-cancer screening before and after transplant is essential (PMID: 38020562).

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12. Treatment

There is **no SFTPA2-specific or curative therapy.** Management is supportive and disease-modifying:

- **Antifibrotics (standard of care for IPF-pattern disease): pirfenidone** and **nintedanib** (NCIT: Pirfenidone, Nintedanib) slow lung-function decline but do not reverse fibrosis (PMID: 38263193; PMID: 38805909).
- **Immunomodulation (surfactant-gene–specific consideration):** Surfactant-gene ILD *"may improve with steroids, hydroxychloroquine or azithromycin"* (PMID: 32437840) — a notable contrast to sporadic IPF, where immunosuppression is generally avoided.
- **Lung transplantation:** A valid option in adults with surfactant-disorder ILD, with good reported prognosis; immunosuppressive protocol was unchanged but close lung-cancer screening is required before and after (PMID: 38020562).
- **Lung-cancer management/surveillance:** CT screening given high cancer risk in SRG carriers (PMID: 38575158).
- **Supportive care:** Supplemental oxygen, pulmonary rehabilitation, vaccination, treatment of PAH and GERD.
- **Experimental / mechanistically rational agents (preclinical IPF):** Selective **Src-kinase inhibition** (saracatinib) (PMID: 35998281), selective **PDE4 inhibitors** (PMID: 38805909), **PD-1/PD-L1 checkpoint** modulation (PMID: 38263193), proton-pump-inhibitor/pirfenidone combinations (PMID: 36450789), and anti–miR-21 strategies (PMID: 33394826) — none validated specifically for ILD2.

Mechanistically, the UPR→TGF-β1→EMT axis suggests TGF-β pathway inhibition and proteostasis/ER-stress modulation as rational targets; in vitro, pan-TGF-β antibody, TGF-β receptor antagonist, and LTBP silencing reversed mutant-SP-A2–induced EMT and cell death (PMID: 23223528).

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13. Prevention

- **Primary prevention:** No means to prevent the inherited mutation; **smoking cessation** and avoidance of fibrogenic/carcinogenic inhalational exposures are prudent in carriers.
- **Secondary prevention: Cascade genetic screening** of relatives, longitudinal PFT/HRCT monitoring of carriers, and **CT lung-cancer surveillance**.

- **Tertiary prevention:** Antifibrotics, pulmonary rehabilitation, oxygen, PAH management, timely transplant referral.
- **Genetic counseling:** Essential for autosomal-dominant, incompletely penetrant, cancer-predisposing disease; includes reproductive options (prenatal/preimplantation testing) and family planning guidance.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human SP-A is encoded by **two genes, SFTPA1 and SFTPA2**, whereas **rodents have a single Sftpa gene** (PMID: 34484180). This architectural difference complicates direct modeling of SFTPA2-specific disease.
- **Natural disease in animals:** No well-established naturally occurring SFTPA2 orthologous Mendelian ILD is documented here; this is a knowledge gap for OMIA/veterinary resources.
- **Comparative biology:** SP-A's innate-immune/alveolar-macrophage function is conserved, but the two-gene human system means human SP-A1/SP-A2 variant biology is not fully recapitulated by single-gene rodents.

15. Model Organisms

- **In vitro / cellular models (primary evidence base):** Human **lung epithelial cell lines** expressing mutant SP-A2 reproduce ER retention, UPR activation, latent TGF-β1 secretion, EMT and cell death (PMID: 23223528); recombinant-protein secretion assays demonstrate abolished secretion (PMID: 19100526; PMID: 32855221).
- **Mouse models:** Humanized SP-A1/SP-A2 variant mice have been used to study differential effects on alveolar macrophages, surfactant structure/function, and survival after infection/oxidative stress (PMID: 34484180; PMID: 32670284).
- **Limitations:** Single rodent Sftpa gene; models emphasize innate-immune/surfactant function rather than full UIP fibrosis + lung-cancer phenotype. No model fully recapitulates the complete human ILD2 syndrome.

Mechanistic Model / Interpretation

ILD2 is best understood as a **proteostasis/gain-of-function disease of the alveolar type II cell**. A single CRD missense change in SFTPA2 is sufficient, in a dominant fashion, to misfold SP-A2 and trap it in the ER. Crucially, this is **not haploinsufficiency**: protein is made but not secreted, and it is the *retained, misfolded protein* that is pathogenic. The retained protein's ER burden triggers the UPR, and the UPR — in a manner specific to these mutants, not to generic misfolded proteins — licenses secretion of latent TGF-β1 together with its latency-binding proteins LTBP-1/4. TGF-β1 then acts as the central fibrogenic effector, driving EMT and epithelial death, which through repetitive injury and myofibroblast activation yields UIP-pattern fibrosis. A parallel branch — epithelial dysregulation and bronchiolar metaplasia — explains the signature lung-adenocarcinoma predisposition. This model is strongly supported because each step was experimentally demonstrated and the downstream EMT/death was *reversible* by blocking TGF-β or LTBPs.

Upstream → Downstream	Evidence strength
SFTPA2 CRD mutation → misfolding	Strong (structural prediction + recombinant assays)
Misfolding → ER retention / no secretion	Strong (recombinant + 11-variant cohort)
ER retention → UPR	Strong (required for maximal TGF-β1 output)
UPR → latent TGF-β1 (+LTBP-1/4)	Strong, mutant-specific
TGF-β1 → EMT + cell death	Strong (reversible by blockade)
EMT/injury → UIP fibrosis	Inferred (integration + patient histology)
Epithelial dysregulation → lung adenocarcinoma	Epidemiologic/clinical

Evidence Base

PMID	Role in this report
19100526	**Foundational** — links SFTPA2 CRD missense (G231V, F198S) to familial fibrosis + lung cancer; ER retention, no secretion
23223528	**Mechanism** — mutant SP-A2 → UPR → latent TGF-β1 → EMT; reversible by TGF-β/LTBP blockade
32855221	11 SFTPA1/SFTPA2 mutations: production preserved, secretion abolished; 93% family ILD/adenocarcinoma
32437840	Imaging (ground-glass + cysts) and immunomodulatory treatment response
36370864	SRG-mutation ILD: younger, lower FVC/DLCO, median survival ~44 months
38575158	High lung-cancer risk in SRG carriers → CT screening
36135709	SFTPA2 variants ~1.3% of IPF; may occur without cancer/family history; longer telomeres
34484180	SP-A2 innate-immune/alveolar-macrophage role; human two-gene architecture
22853774	Familial vs sporadic PF: younger onset (61.4 vs 66.6 y), UIP histology
38020562	Lung transplant outcomes in surfactant-disorder ILD; cancer screening
42510961	NGS in PF; TGF-β/Wnt/PI3K-Akt pathways
35795226	IPF molecular genetics, prevalence, epigenetics
41618408	Korean cohort; population-specific architecture; familial faster decline
36833253	NGS supports diagnosis of familial interstitial pneumonia

Database annotations: OMIM 178500; Monarch/HPO (12 ILD2 phenotype annotations); ClinVar (137 SFTPA2 records; pathogenic cluster in CRD); UniProt Q8IWL1 / InterPro IPR033990, IPR001304 / Pfam PF00059 (collectin C-type lectin CRD; secreted + lamellar body localization).

Limitations and Knowledge Gaps

1. **Final fibrosis step is inferential.** The jump from demonstrated epithelial EMT/death to full UIP tissue remodeling integrates cell-based data with patient histology rather than a single demonstrated in vivo chain.
2. **Cancer-mechanism gap.** The link between SFTPA2 mutation and lung adenocarcinoma is epidemiologic/clinical; the molecular route (via bronchiolar metaplasia/epithelial dysregulation) is not mechanistically resolved.
3. **No faithful whole-animal model.** The human two-gene SP-A system is not recapitulated by single-Sftpa rodents; no model reproduces the complete fibrosis + cancer phenotype.
4. **Penetrance/modifiers unquantified.** Incomplete, age-dependent penetrance implies modifiers and GxE effects that remain uncharacterized for SFTPA2 specifically.
5. **Epidemiology is coarse.** ILD2-specific prevalence/incidence, sex ratio, and QoL metrics are not precisely established; figures are extrapolated from broader IPF/SRG cohorts.
6. **VUS burden.** A large fraction of ClinVar SFTPA2 records are VUS, complicating clinical interpretation.

Proposed Follow-up Experiments / Actions

1. **AT2-specific / organoid models:** Generate patient-iPSC–derived AT2 organoids or humanized SFTPA1/SFTPA2 knock-in mice carrying p.Gly231Val/p.Phe198Ser to test whether the UPR→TGF-β1→EMT axis produces bona fide progressive fibrosis in vivo.
2. **Therapeutic proof-of-concept:** Test ER-stress/UPR modulators (e.g., chemical chaperones, PERK/IRE1 modulators) and TGF-β pathway inhibitors in these models, building on the demonstrated reversibility of EMT by TGF-β/LTBP blockade.
3. **Cancer-mechanism studies:** Lineage-trace bronchiolar metaplastic cells in carriers to dissect the fibrosis-to-adenocarcinoma transition and identify early biomarkers.
4. **Variant curation:** Apply functional secretion assays systematically to reclassify SFTPA2 VUS, building a ClinGen-style evidence framework.
5. **Natural-history registry:** Assemble an SFTPA2-specific cohort to quantify penetrance, decline rates, sex distribution, QoL, and transplant/cancer outcomes.

6. **Clinical protocol:** Formalize carrier management — cascade testing, antifibrotic timing, trial of immunomodulation in surfactant-gene cases, and structured CT lung-cancer surveillance.
